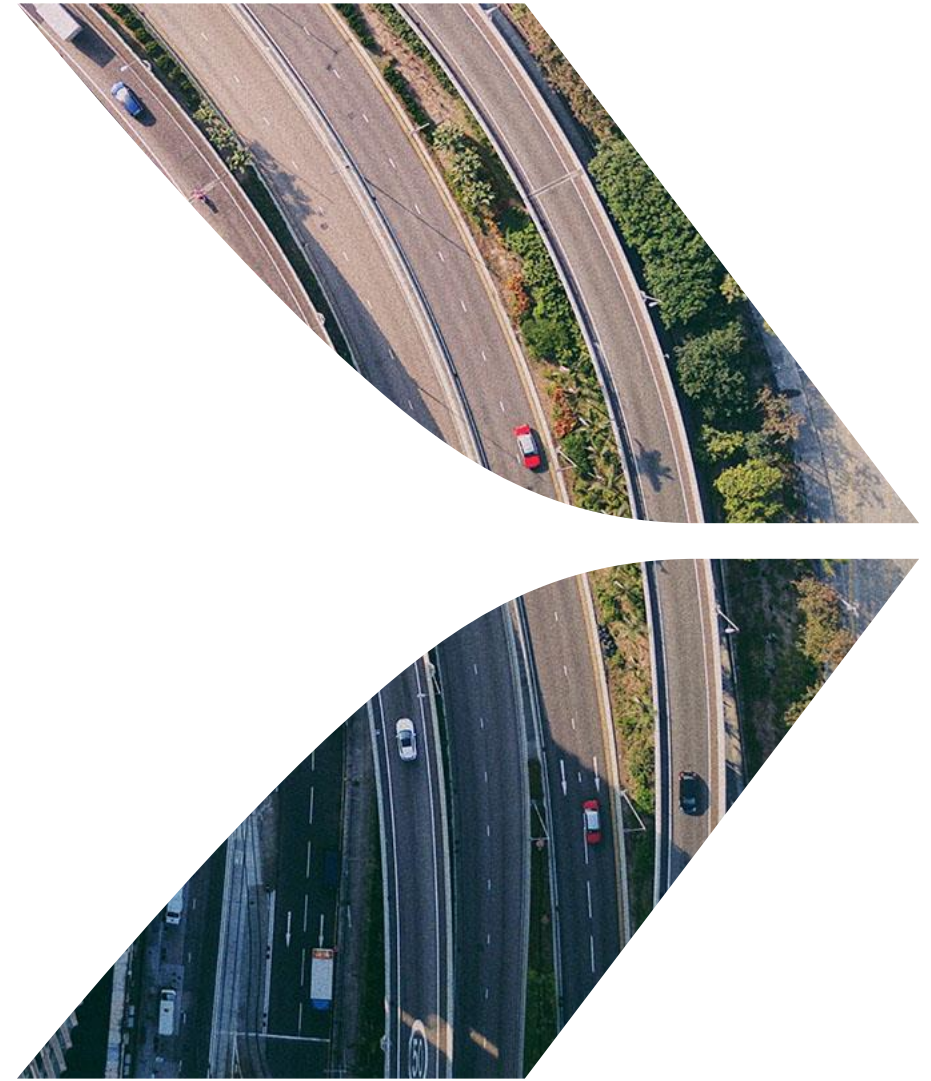


AUGUST 19, 2026

Stochastic simulation & derivatives risk engine

Berkeley Haas MFE Industry Project | Kickoff



Who is Capitolis

Creating safer capital markets

The challenge

- Financial institutions need to continuously optimize financial resources to serve clients, support growth, and produce shareholder value.
- Increasing regulatory demands on capital and liquidity drive the need for innovative solutions.

Two means to address the challenge

- **Capital Marketplace:** Identify and access new pools of capital from institutional investors to support the provision of core financial services by banks in a safe and efficient manner. Capitolis has multiple product offerings including a product that is an SEC-registered broker-dealer and securities-based swap dealer (SBSD) with an OTC derivatives trading desk.
- **Portfolio Optimization:** Optimize the existing portfolio without changing market risk by reducing or novating positions through compression or other optimization exercises with multiple counterparties.

Capital Marketplace and Efficient Swap Financing

Bringing banks and investors together on a single platform

Capital Marketplace

- Live with 18 bank clients, 70+ investors, and \$35.4B in assets on platform. Products span Efficient Swap Financing (ESF), Unsecured Bond Borrow, Securities Financing Platform, and Counterparty Risk Financing.

Efficient Swap Financing (ESF)

- A platform providing balance sheet efficiencies to bank Total Return Swap (TRS) and other derivative counterparties by uniquely uniting funded investors and banks.
- Transforms illiquid bilateral transactions into liquid CUSIP instruments, providing banks with derivative capacity uncorrelated to traditional markets.

Why ESF drives our risk work

- ESF creates a portfolio of OTC derivative positions (equity and bond TRS, bond forwards, etc.) that must be priced, risk-managed, and capitalized. **This is the portfolio you will be simulating in this project.**

The Capitolis team

We are your professional supervisors

Jed Serby

Director, Model Risk

I serve as the lead for quantitative analytics and internal model development within the Risk department here at Capitolis, supporting both Capital Marketplace and Portfolio Optimization products.

Prior to joining Capitolis in November 2025, I led a commercial credit risk model development team at CapitalOne in New York. Prior to this, I served as the Head of Model Development Quality Assurance at U.S. Bank, and as a quantitative consultant at Promontory Financial Group.

I hold an MFE from Berkeley, an MA in actuarial science from Columbia University, and a B.A. in Economics and Mathematics from Emory University.

Frank Butler

Senior Risk Specialist

I provide quantitative support focused on model development and validation within the Risk Department for the Capital Marketplace.

I bring nearly 35 years of financial services experience to the Firm, most recently serving as a Managing Director at FMS Wertmanagement Service GmbH, where I developed new financial products. Prior to this, I held senior quantitative analytics roles at HypoVereinsbank, Barclays Bank, and Fuji Capital Markets. Before entering the financial services industry, I worked as a staff physicist at the IBM T.J. Watson Research Center, performing supercomputing calculations of the properties of hadronic matter.

I hold a Ph.D. in theoretical physics from Columbia University and an A.B. in physics from the University of Chicago.

Paul Tatulea

Algo Engineer

I've spent the past seven years working on risk and optimization engines for portfolio optimization. My work spans rates derivatives (interest rate swaps, cross-currency swaps, swaptions, cap/floors, and CMS spread options), FX products (forwards and options), and credit products (credit default swaps), with a focus on risk sensitivities, XVA, and regulatory capital frameworks like SIMM and SA-CCR. Across the industry, the products I work on have compressed trillions of dollars in notional and delivered billions in initial margin savings. I am passionate about financial markets, derivatives pricing and risk, and scaling systems for high throughput and reliability. But most of all, I enjoy learning from and working with similarly passionate people.

Project objectives

Build the computational backbone of a counterparty credit risk framework

Design and implement a modular Monte Carlo simulation engine capable of ingesting OTC derivative pricing models, market data, and cross-asset correlation structures.

1. Stochastic models

Implement risk factor models (your choice) and simulate correlated paths. You must be ready to defend your decisions.

2. Pricing model integration

Feed simulated market states into the provided pricing models to generate scenario NPVs.

3. Exposure and Greeks

Derive exposure profiles (PFE, MPE, EE) and efficient Greeks computation across scenarios. *Extra Credit Opportunity: xVA*

4. Performance

Optimize engine for speed and scalability. Using cutting edge techniques to make this fast.

Materials provided vs. your responsibilities

What Capitolis provides

16 trades across 4 instrument types for 3 counterparties:

- 8 Equity TRS (incl. 1 JPY compo trade)
- 4 Bond Forwards
- 4 Bond TRS
- Underlying assets:
 - 41 underlying equity positions across baskets
 - 5 bond types

Risk factors per trade:

- USD Equity TRS: USD curve + equity spots
- JPY Compo TRS: USD curve + equity spots + USDJPY FX
- Bond Fwd/TRS: USD curve only (risk-free bonds)

Capitolis is strictly seller on these transactions!

What you collect and build

Market data:

- USD OIS (SOFR) curve
- Equity spots and dividend rates per name
- USDJPY FX spot and forward curve
- Volatilities for all risk factors
- Pairwise correlation matrix
- See more requirements in the MARKET_DATA.md

The Risk Engine:

- Stochastic models for rates, equity, FX
- Correlated path simulation
- Exposure aggregation by netting set, along with Greeks
- *Extra Credit: source and incorporate risky bonds in example trades*

Pricing Models

We are providing a standalone Python pricing library for your use

Folder structure of the capitolis_pricers Zip File

- trade_data/ Capitolis-supplied trade CSVs
- trade_data/underlyings/ Reference assets for the trade mapping on equities (by trade_id) and bonds (by bond_id)
- capitolis_pricers/ Pricing modules: equity_trs.py, bond_trs.py, bond_forward.py, plus curve and market builders
- capitolis_pricers/examples/ price_all.py (runs full book)
- Full documentation of every field, pricer contract, and data layer: **README.md**

The pricing model connection

- Every pricer is a pure function of a MarketState.
- To build exposure, your engine simulates risk factors, assembles a MarketState per (scenario, horizon) node, and calls pricer.npv(node_market).
- This is the integration point between your simulation and the provided pricing models.

Exposure and PFE

What is exposure?

- When a counterparty defaults on an OTC derivative, your exposure is the positive mark-to-market at that moment, net of collateral collected. Because you cannot know when default will occur or what rates and prices will be, monte carlo is one method to derive hypothetical market scenarios across time and measure the distribution of potential exposures at a given horizon.

Key exposure metrics: We require the 10-day close out windows over one year horizon.

- **Expected Exposure (EE):** The mean of positive NPVs across scenarios over the 10-day close out window at each time step.
- **Potential Future Exposure (PFE):** We require the 99th percentile for 10-day close out windows over one year horizon. This would be calculated by the 10-day period at a given time step for each time step in the one-year horizon.
- **Maximum PFE (MPE):** The peak of the profile of 10-day PFEs across all time horizons.
- Exposures are computed per trade and aggregated up to the counterparty level.

Key assumptions

- Assume no initial margin.
- Assume collected/posted variation margin on a given date in the simulation is the NPV of the trade on the prior day on the path. Because the counterparty defaults, variation margin discontinues over the close out period and exposure is how much it moves from the NPV on t-1.
- Use a 10-day close out period. That means, assume it takes 10 days from default to close out the trade – what does the exposure grow to over that period? How you get 10-day windows and cover all days over the one-year horizon is up to you (interpolation?)
- The number of simulations you incorporate is up to you but must evidence convergence
- General rule of thumb: as we are strictly sellers on derivatives, *falling* market values increase counterparty exposure to us – should they default it is not recoverable – and thus higher PFEs and MPEs all else equal

Project timeline

August 19 - October 3 | ~7 weeks | Weekly check-ins

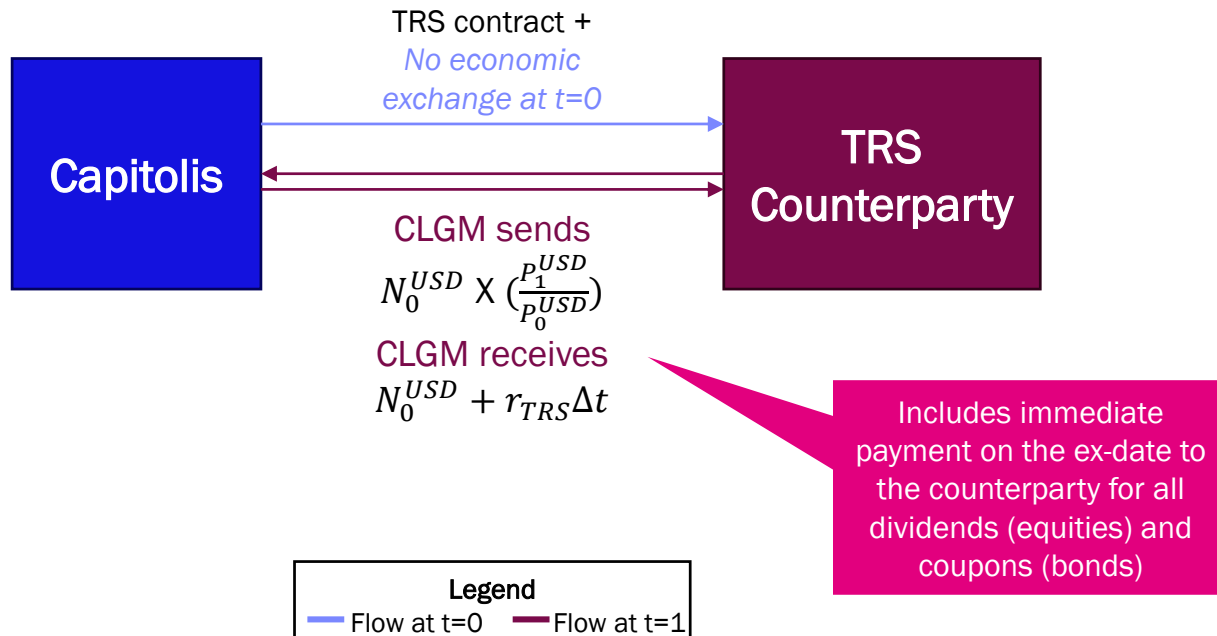
Stage	Description	Deliverables	Aug 24	Aug 31	Sept 7	Sept 14	Sept 21	Sept 28	Oct 5
Onboarding and Data Collection	Review provided pricers and trade data. Source market data (curves, spots, FX, dividends). Validate pricing against sample market. Set up GitHub repo and project structure.	- GitHub Repo - Data - Pricers and trades (Capitolis)							
Foundations Engine Buildout	Choose and implement stochastic models for rates, equity, FX. Build correlation proxy method from historical data. Stand up the full Monte Carlo engine with joint simulation of correlated risk factors. Integrate with provided pricers to generate scenario NPVs. Compute netting-set level exposure profiles. Demonstrate speed and performance	- Risk Engine with outputs - Benchmarks for speed - Analytical benchmarks for accuracy							
Analytics	Implement Greeks (bumped or pathwise)	Greeks							
Documentation and final presentation	Finalize technical report with benchmarks. Presentation to Capitolis Risk department	- Codebase in GitHub - Technical report (details in appendix)							

Extra credit options not put in swimlanes:

- Risk bonds + CDS data for new sample trades
- xVA outputs

Appendices

TRS: Passing on all gains for equities (and bonds!)

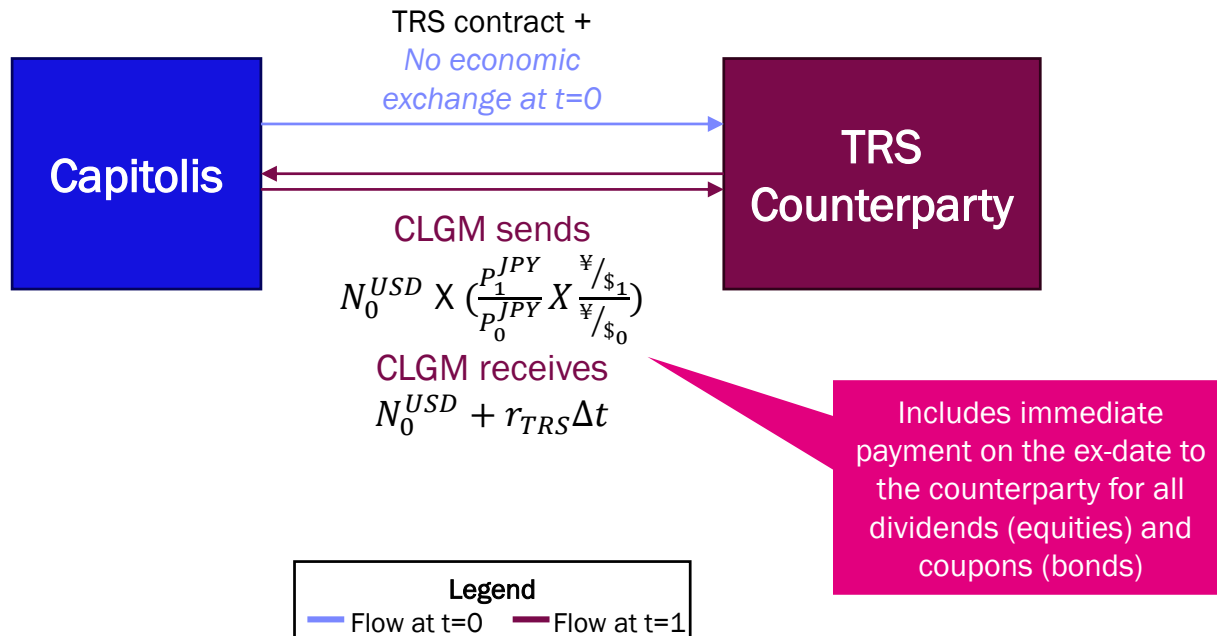


Example: Price ↑10%

Metric	Symbol	t=0	t=1
Notional	N^{USD}	\$100	\$100
Eq. Price	P^{USD}	\$10	\$11
Shares	Z	10	10
Equity Market Value in USD	$N_0^{USD} \times \left(\frac{P_1^{USD}}{P_0^{USD}}\right)$	\$100	\$100 X (11/10) = \$ 110
CLGM sends to TRS Counterparty	$N_0^{USD} \times \left(\frac{P_1^{USD}}{P_0^{USD}}\right)$	-	\$100 X (11/10) = \$ 110
CLGM gets from TRS Counterparty	$N_0^{USD} + r_{TRS}\Delta t$	-	\$100 + funding leg

Compo: TRS Performance Leg in USD, Funding Leg in USD

Counterparty takes FX risk as return is a function of two risk factors



Example: Price ↑1% & JPY/USD ↓ 5%

Metric	Symbol	t=0	t=1
JPY/USD	¥/\$	50 bp	47.5 bp
Notional	N^{USD}	\$100	\$100
Eq. Price	P^{JPY}	¥50	¥50.50
Shares	Z	400	400
CLGM sends to TRS Counterparty	$N_0^{USD} \times \left(\frac{P_1^{JPY}}{P_0^{JPY}} \times \frac{\text{¥}/\$1}{\text{¥}/\$0} \right)$	-	\$100 X (47.5/50 X ¥50.50/¥50) = \$ 95.95
CLGM gets from TRS Counterparty	$N_0^{USD} + r_{TRS} \Delta t$	-	\$100 + funding leg

Bond forward: agreeing today on a future bond price

The least complex derivative in the portfolio as a function solely of interest rates

What it is?

- A contract to buy (long) or sell (short) a bond at a future date at an agreed clean price. No cash changes hands at inception. At settlement, the buyer pays the strike and receives the bond. Capitolis is always the seller so falling prices increase the counterparty's exposure.

$$NPV_{for\ buyer} = \frac{notional}{par} X (Forward\ Value_0^T - Strike_0^T) X df^{T \rightarrow 0}$$

Risk factors

- Bond forwards in this portfolio are risk-free (government/agency) bonds discounted on the USD curve.
- The only risk factor is the USD short rate. This makes them a clean test case: you can validate your rate simulation against analytical bond forward prices before adding equity and FX complexity.
- *Extra credit*: if you add issuer credit (CDS curves), the bond prices credit-risky and the forward picks up credit spread as an additional risk factor. Create trades to add this risk.

Exposure and Regulatory Requirements

The regulatory purpose of the engine you are building

SA-CCR and bank regulatory capital

- SA-CCR (Standardized Approach for Counterparty Credit Risk) is the Basel III framework banks use to convert exposure into a capital charge:

Exposure at Default (EAD) = $\alpha \times (\text{Replacement Cost} + \text{PFE add-on})$.

- Banks must hold capital against this EAD.
- This is what drives demand for Capitolis products — CM helps banks optimize these capital charges.

SEC Rule 18a-1 for SBSDs

- As a registered SBSD, Capitolis computes net capital under SEC Rule 18a-1.
- Rule 18a-1 requires specific market risk and **credit risk charges** on OTC derivative positions.
- Internal models that produce VaR and **MPE** feed directly into these capital computations.

Final Deliverable Descriptions

Simulation Engine Codebase

- Well-documented Python implementation with modular architecture enabling extensions to additional asset classes or model types

Technical Report

- Documentation of model calibration methodology against provided market data, with benchmarking of simulated prices against analytical solutions where available.
- The paper should include technical write-up covering model choices, numerical methods, key results, and suggestions for production hardening.
- The report should include performance analysis demonstrating computational capabilities, including convergence and optimization metrics

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